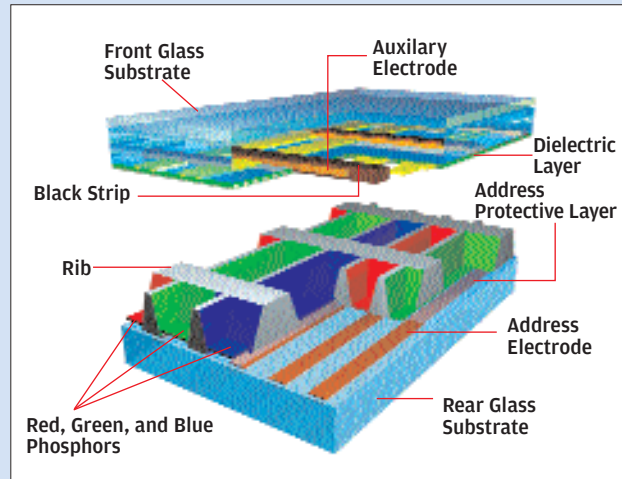


Plasma Displays—More CRT than LCD

The working of an LCD system should be clear now. But what about plasma TVs? Plasma is another flat screen display type that most people confuse with LCDs. However, the extremely flat (read compact) dimensions is all that these two technologies have in common.

A plasma TV is comprised of two sheets of glass with a series of corrugations that have coloured phosphors in between. All this is on the bottom glass layer. The top glass layer is embedded with the electrodes, and both these layers meet to form hundreds of thousands of tiny pixels, each of which is sealed with a gas inside it. Gases that have been used in plasmas are typically neon and argon, and more recently, xenon has also been used. All these pixels are naturally in the same plane, that is, parallel to each other. A single pixel is made up of three sub-pixels, one each of red, green, and blue. Note that these are the colours of the phosphor sandwiched between the two glass sheets.

Now each of the phosphors can be individually controlled as they have their own electrodes. When active, the electrodes bring the rare gases to a plasma state, where they can conduct electricity. The gases react with the phosphor to emit a burst of ultraviolet (UV) light that is invisible to the human eye. The energy from this UV light stimulates the colour phosphors, making them glow. A high voltage is supplied to activate the gases, and after that a much lower voltage supplies them with the necessary stimulation to keep the pixels in the On position.



One of the biggest acclaimed disadvantages of plasma is supposed to be the display's abysmal life. However this is *not* true. A typical plasma display lasts for around 25,000 hours, which incidentally is also the life of a regular CRT monitor. Think of it this way: even if you use your plasma display for 12 hours a day, every day, it will last you for $25,000 / 12 = 2,083$ days, or roughly 5 years and 8 months!

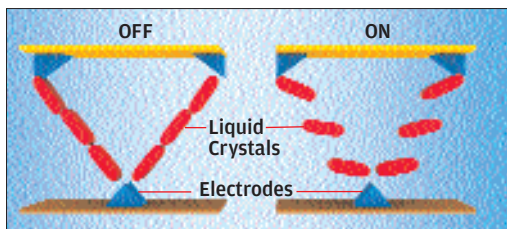


Figure 3: Liquid crystals in an MVA panel

MVA panels attempt to address the shortcomings inherent in TN and IPS panels

How MVA Works

As the name implies, MVA (Multi-Domain Vertical Alignment) panels see each pixel assigned a domain, or an area all to itself. Now these domains work in synchrony with each other. As you can see in Figure 3, the crystals form a sort of a shutter system whereby if one domain allows light to pass through (in the On position), the neighbouring domain will have its crystals aligned at a certain angle to shut out the light, or deflect it. If the monitor is to display white, you see all the crystals aligned somewhat parallel to the plane surface.

PVA

PVA (Patterned Vertical Alignment) was developed by Samsung as an offshoot of the technology used in MVA panels. Some people even criticise the technology stating there's enough in common to conclude that Samsung developed PVA to avoid paying licensing fees to Fujitsu for bulk-producing these panels.

However, a closer look at PVA as a panel technology will suggest differences that lead us to accept PVA as an independent (if not innovative) product that owes its origins to MVA.

In PVA panels, the alignment of crystals is identical to MVA panels. The domain principle is

also used. This allows PVA panels to enjoy amazing viewing angles similar to MVA panels. The disadvantage of a slow pixel response times is also present in PVA matrices. A significant advantage is the improved contrast ratios. PVA panels can actually achieve a contrast ratio of up to 1000:1 in real figures, not what manufacturers claim. This is why PVA panels are finding their way into almost all LCD TVs available in the international market today. The fact that Samsung is the only player manufacturing PVA matrices also means that quality of such panels will be more or less identical.

PVA matrices can be found in the 19-inch-and-above categories. A lot of the massive LCD TVs in the range of 65+ inches (and even the smaller 26+ inches and above) are based on PVA matrices.

The Future

LCDs aren't the future. They are very much the present. As of today, CRT is a dead technology, and manufacturers and consumers alike are focusing on the flat panel market. The binaries of breakthrough technological advances and sharply declining costs is the culprit for the ever increasing popularity of LCD monitors across the globe.

The best news for you, our readers, is the conscious move towards larger LCD panels by the bigger manufacturers. Two of the leading flat panel manufacturers, LG-Philips and Samsung, each with an OEM market share of 22 per cent, are pushing out larger panels to fuel the sudden demand for more desktop space and larger screens for home entertainment. This drives the cost of mainstream panels down, and the trend of falling prices should continue in this product segment through this year and beyond. ■

michael_browne@thinkdigit.com